



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program:Bachelor of Technology
Semester : III

Branch:Computer Science and Engineering
Specialization: Computer Science and Engineering

Session: 2021

S.No.	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical			L	T	P	
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem	Term work					
							Lab Work & Sessional					
1.	CH-31	Energy & Environmental Engineering	70	20	10	-	-	100	3	1	-	4
2.	MA-31	Mathematics III	70	20	10	-	-	100	3	1	-	4
3.	CS-31	Data Structure	70	20	10	30	20	150	3	-	2	4
4.	CS-32	Digital Systems	70	20	10	30	20	150	3	-	2	4
5.	CS-33	Object Oriented Programming & Methodology	70	20	10	30	20	150	3	-	2	4
6.	CS-34	Computer programming python	-	-	-	30	20	50	-	-	4	2
7.	CS-35	Seminar / GD	-	-	-	-	50	50			4	2
		Total	350	100	50	120	130	750	15	2	14	24

***End Semester Examination will be of three hours.**

PRACTICALS SUBJECT CS- 31:Data structure
CS-33: OOPM

CS-32:DigitalSystem
CS-34:Computerprogramming

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

SYLLABUS OF COURSES TO BE OFFERED “Bachelor of Technology”

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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CH-31 Energy & Environmental Engineering

UNIT 1: Introduction to Energy Science:

Introduction to energy systems and resources; Introduction to Energy, sustainability & the environment; Overview of energy systems, sources, transformations, efficiency, and storage; Fossil fuels (coal, oil, oil-bearing shale and sands, coal gasification) - past, present & future, Remedies & alternatives for fossil fuels - biomass, wind, solar, nuclear, wave, tidal and hydrogen; Sustainability and environmental trade-offs of different energy systems; possibilities for energy storage or regeneration (Ex. Pumped storage hydro power projects, superconductor-based energy storages, high efficiency batteries)

UNIT 2: Ecosystems

Concept of an ecosystem; Structure and function of an ecosystem; Producers, consumers and decomposers; Energy flow in the ecosystem; Ecological succession; Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of the following ecosystem (a.) Forest ecosystem (b) Grassland ecosystem (c) Desert ecosystem (d) Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

UNIT 3: Biodiversity and its conservation

Introduction – Definition: genetic, species and ecosystem diversity; Bio-geographical classification of India; Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values; Biodiversity at global, National and local levels; India as a mega-diversity nation; Hot-spots of biodiversity; Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; Endangered and endemic species of India; Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT 4: Environmental Pollution

Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards; Solid waste Management: Causes, effects and control measures of urban and industrial wastes; Role of an individual in prevention of pollution; Pollution case studies; Disaster management: floods, earthquake, cyclone and landslides.

UNIT 5: Social Issues and the Environment

From Unsustainable to Sustainable development; Urban problems related to energy; Water conservation, rain water harvesting, watershed management; Resettlement and rehabilitation of people; its problems and concerns. Case Studies

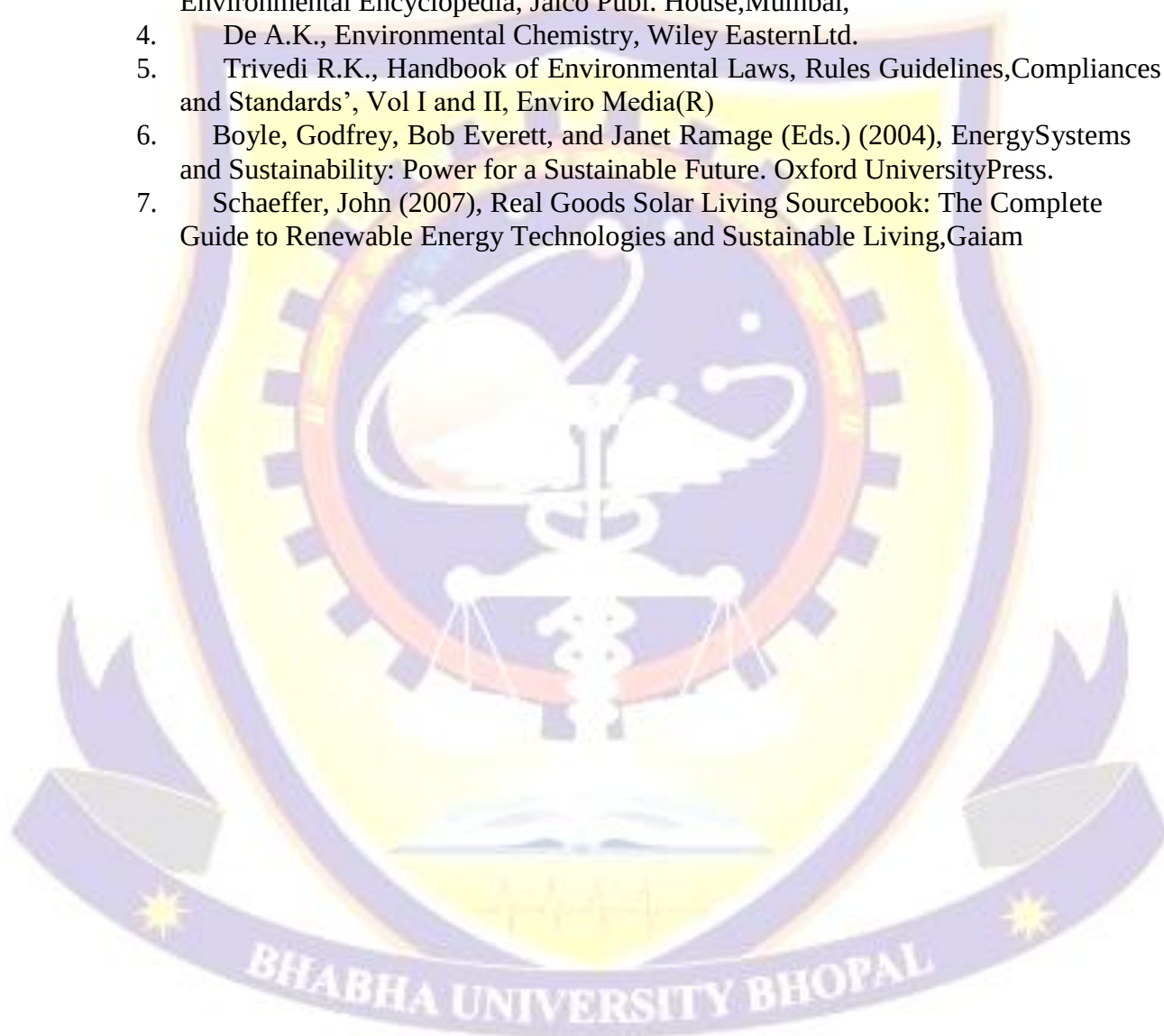
Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies Wasteland reclamation; Consumerism and waste products; Environment Protection Act; Air (Prevention and Control of Pollution) Act; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act; Issues involved in enforcement of environmental legislation;



Publicawareness.

REFERENCES

1. Brunner R.C., 1989, Hazardous Waste Incineration, McGraw Hill Inc.
2. Clark R.S., Marine Pollution, Clanderson Press Oxford(TB).
3. Cunningham, W.P. Cooper, T.H. Gorhani, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumbai,
4. De A.K., Environmental Chemistry, Wiley Eastern Ltd.
5. Trivedi R.K., Handbook of Environmental Laws, Rules Guidelines, Compliances and Standards', Vol I and II, Enviro Media(R)
6. Boyle, Godfrey, Bob Everett, and Janet Ramage (Eds.) (2004), Energy Systems and Sustainability: Power for a Sustainable Future. Oxford University Press.
7. Schaeffer, John (2007), Real Goods Solar Living Sourcebook: The Complete Guide to Renewable Energy Technologies and Sustainable Living, Gaia





MA-31 Mathematics III

Unit 1: Numerical Methods – 1: Solution of polynomial and transcendental equations – Newton-Raphson method and Regula-Falsi method. Finite differences, Relation between operators, Interpolation using Newton's forward and backward difference formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange's formulae.

Unit 2: Numerical Methods – 2: Numerical Differentiation, Numerical integration: Trapezoidal rule and Simpson's 1/3rd and 3/8 rules. Solution of Simultaneous Linear Algebraic Equations by Gauss's Elimination, Gauss's Jordan, Crout's methods, Jacobi's, Gauss-Seidel,

Unit 3: Numerical Methods – 3: Ordinary differential equations: Taylor's series, Euler and modified Euler's methods. Runge-Kutta method of fourth order for solving first and second order equations. Milne's and Adam's predictor-corrector methods. Partial differential equations: Finite difference solution two dimensional Laplace equation and Poisson equation, Implicit and explicit methods for one dimensional heat equation (Bender-Schmidt and Crank-Nicholson methods)

Unit 4: Transform Calculus: Laplace Transform, Properties of Laplace Transform, Laplace transform of periodic functions. Finding inverse Laplace transform by different methods, convolution theorem. Evaluation of integrals by Laplace transform, solving ODEs by Laplace Transform method, Fourier transforms.

Unit 5: Concept of Probability: Probability Mass function, Probability Density Function, Discrete Distribution: Binomial, Poisson's, Continuous Distribution: Normal Distribution, Exponential Distribution.

REFERENCE

1. P. Kandasamy, K. Thilagavathy, K. Gunavathi, Numerical Methods, S. Chand & Company, 2nd Edition, Reprint 2012.
2. S.S. Sastry, Introductory methods of numerical analysis, PHI, 4th Edition, 2005.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 35th Edition, 2010.
5. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010.
6. Veerarajan T., Engineering Mathematics, Tata McGraw-Hill, New Delhi, 2008.
7. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003 (Reprint).
8. S. Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
9. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968. Statistic



CS-31 Data Structure

Unit -1:-Review of C programming language. Introduction to Data Structure: Concepts of Data and Information, Classification of Data structures, Abstract Data Types, Implementation aspects: Memory representation. Data structures operations and its cost estimation. Introduction to linear data structures- Arrays, Linked List: Representation of linked list in memory, different implementation of linked list. Circular linked list, doubly linked list, etc.

Unit -2:-Stacks: Stacks as ADT, Different implementation of stack, multiple stacks. Application of Stack: Conversion of infix to postfix notation using stack, evaluation of postfix expression, - Queues: Queues as ADT, Different implementation of queue, Circular queue, Concept of Dqueue and Priority Queue, , Application of queues.

Unit-3:-Tree: Definitions - Height, depth, order, degree etc. Binary Search Tree - Operations, Traversal, Search. AVL Tree, Heap, Applications and comparison of various types of tree; Introduction to forest, multi-way Tree, B tree, B+ tree, B* tree.

Unit-4:-Graphs: Introduction, Classification of graph: Directed and Undirected graphs, etc, Representation, Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS), Graph algorithm: Minimum Spanning Tree (MST)- Kruskal, Prim's algorithms. Dijkstra's shortest path algorithm; Comparison between different graph algorithms. Application of graphs.

Unit-5:-Sorting: Introduction, Sort methods like: Bubble Sort, Quick sort. Selection sort, Heap sort, Insertion sort, Shell sort, Merge sort and Radix sort; comparison of various sorting techniques. Searching: Basic Search Techniques: Sequential search, Binary search, Comparison of search methods

LIST OF EXPERIMENTS

1. Write a program to search an element in the array using Linear and Binary Search.
2. Write a program to perform the following operation in Matrix:
 1. Addition 2. Subtraction 3. Multiplication 4. Transpose
3. Write program for implementing the following sorting methods to arrange a list of integers in ascending order:
 - a) Quick sort b) Selection sort c) Insertion sort d) Merge sort
4. Write a program that uses stack operations to convert a given infix expression into its postfix equivalent.
5. Write a program to merge two sorted array into one sorted array.
6. Write a program to implement stack using array and linked list.
7. Write a program to implement queue and circular queue using array.
8. Write a program to insert an element in the beginning and end of singly linked list.
9. Write a program to insert an element at any position in singly and doubly linked list.
10. Insert and delete a node at any position in doubly linked list.



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REFERENCES

1. Aho, Hopcroft, Ullman, “Data Structures and Algorithms”, Pearson Education.
2. N. Wirth, “Algorithms + Data Structure = Programs”, Prentice Hall.
3. Jean – Paul Trembly , Paul Sorenson, “An Introduction to Structure with application”, TMH.
4. Richard, Gilberg Behrouz, Forouzan , “Data structure – A Pseudocode Approach with C”, Thomson press.
5. AM Tanenbaum, Y Langsam & MJ Augustein, “Data structure using C and C++”, Prentice Hall India.
6. Robert Kruse, Bruce Leung, “Data structures & Program Design in C”, Pearson Education.





CS-32 Digital Systems

Unit 1: Review of number systems and number base conversions. Binary codes, Boolean algebra, Boolean functions, Logic gates. Simplification of Boolean functions, Karnaugh map methods, SOP-POS simplification, NAND-NOR implementation.

Unit 2: Combinational Logic: Half adder, Half subtractor, Full adder, Full subtractor, look-ahead carry generator, BCD adder, Series and parallel addition, Multiplexer – demultiplexer, encoder- decoder, arithmetic circuits, ALU

Unit 3 : Sequential logic: flip flops, D,T, S-R, J-K Master- Slave, racing condition, Edge & Level triggered circuits, Shift registers, Asynchronous and synchronous counters, their types and state diagrams. Semiconductor memories, Introduction to digital ICs 2716, 2732 etc. & their address decoding. Modern trends in semiconductor memories such as DRAM, FLASH RAM etc. Designing with ROM and PLA.

Unit 4: Introduction to A/D & D/A convertors & their types, sample and hold circuits, Voltage to Frequency & Frequency to Voltage conversion. Multivibrators, Bistable, Monostable, Astable, Schmitt trigger, IC 555 & Its applications. TTL, PMOS, CMOS and NMOS logic. Interfacing between TTL to MOS.

Unit 5: Introduction to Digital Communication: Nyquist sampling theorem, time division multiplexing, PCM, quantization error, introduction to BPSK & BFSK modulation schemes. Shannon's theorem for channel capacity.

LIST OF EXPERIMENTS

1. To study and verify the truth tables of various Logic gates
2. To verify the properties of NAND and NOR gates as Universal Building Blocks.
3. Simplification and implementation of a Boolean function
4. Implementation of basic Boolean arithmetic logic circuits such as Half-adder, Half-subtractor, Full adder and Full subtractor
5. Conversion from Binary to Gray and Gray to Binary code
6. To construct a binary multiplier using combinational logic and to verify with the truth table
7. To verify 2-bit Magnitude comparator for all possible conditions
8. Generation of various logical functions using 8-to-1 multiplexer
9. Construction of a 4-bit ripple counter and study of its operation
10. Operation of IC-555 Timer as Monostable, Astable and Bistable multivibrators



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REFERENCES

1. Morris Mano, Digital Circuits & Logic Design, PHI
2. Gothman, Digital Electronics, PHI
3. Tocci, Digital Electronics, PHI
4. Mavino & Leach, Digital Principles & Applications, PHI
5. Taub and schilling, Digital Integrated electronics.
6. Simon Haykin, Introduction to Analog & Digital Communication, Wiley.
7. Lathi B.P., Modern analog & digital communication, Oxford University.





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CS-33 Object Oriented Programming & Methodology

Unit-1:-Introduction to Object Oriented Thinking & Object Oriented Programming: Comparison with Procedural Programming, features of Object oriented paradigm– Merits and demerits of OO methodology; Object model; Elements of OOPS, IO processing.

Unit-2:-Encapsulation and Data Abstraction- Concept of Objects: State, Behavior & Identity of an object; Classes: identifying classes and candidates for Classes Attributes and Services, Access modifiers, Static members of a Class, Instances, Message passing, and Construction and destruction of Objects.

Unit-3:-Relationships – Inheritance: purpose and its types, 'is a' relationship; Association, Aggregation. Concept of interfaces and Abstract classes.

Unit-4:-Polymorphism: Introduction, Method Overriding & Overloading, static and run time Polymorphism.

Unit-5:-Strings, Exceptional handling, Introduction of Multi-threading and Data collections. Case study like: ATM, Library management system.

LIST OF EXPERIMENTS

- 1 Write a program in C++ to find largest no.
- 2 Write a program in C++ to implement the multiplication of [3*3] matrix using Array.
- 3 Write a program in C++ to implement quadratic equation using class & Object.
- 4 Write a program in C++ to implement Copy Constructor.
- 5 Write a program in C++ to implement Inheritance.
- 6 Write a program in C++ to implement Polymorphism.
- 7 Write a program in C++ to implement virtual class & virtual function.
- 8 Write a program in C++ to implement Exception handling.
- 9 Write a program in C++ for function templates.
- 10 Write a program in C++ for writing file operation using file handling.
- 11 To perform the write operation within a file.



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REFERENCES

1. G. Booch, "Object Oriented Analysis& Design", AddisonWesley.
2. James Martin, "Principles of Object Oriented Analysis and Design", Prentice Hall/PTR.
3. Peter Coad and Edward Yourdon, "Object Oriented Design", PrenticeHall/PTR.
4. Herbert Schildt, "Java 2: The Complete Reference", McGraw-Hill Osborne Media, 7th Edition.
5. Timothy Budd, "An Introduction to Object-Oriented Programming", Addison-Wesley Publication, 3rd Edition.
6. Cay S. Horstmann and Gary Cornell, "Core Java: Volume I, Fundamentals", Prentice Hall publication.





CS-34 Computer Programming Practices (PYTHON)

UNIT 1- Introduction:

Basic syntax, Literal Constants, Numbers, Variable and Basic data types, String, Escape Sequences, Operators and Expressions, Evaluation Order, Indentation, Input Output, Functions, Comments.

UNIT 2- Data Structure:

List, Tuples, Dictionary and Sets.

UNIT 3-Control Flow:

Conditional Statements-If, If-else, Nested If-else. Iterative Statement-For, While, Nested Loops. Control statements - Break, Continue, Pass.

UNIT 4-Object oriented programming:

Class and Object, Attributes, Methods, Scopes and Namespaces, Inheritance, Overloading, Hiding and overriding .

UNIT 5-Exception:

Exception Handling, Except clause, Try finally clause, User Defined Exceptions.

LIST OF EXPERIMENTS:

- 1) Python Program to find the area of triangle
- 2) Python program to swap two variables
- 3) Program to print Even Numbers up to N
- 4) Program to check whether the given no. is Prime or not (Method I)
- 5) Write a program to create dataframe for 3 student including name and roll numbers. and add new columns for 5 subjects and 1 column to calculate percentage. It should include random numbers in marks of all subjects
- 6) Recursive Function to Print Multiplication Table
- 7) Recursive Function to print Fibonacci series upto N terms.
- 8) Write a Python program to remove duplicates from a list.

REFERENCES

- Timothy A. Budd: Exploring python, McGraw-Hill Education.
- R. Nageshwar Rao , "Python Programming" , Wiley India ThinkPython:
- Allen B. Downey, O'Reilly Media, Inc.



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CS-35 Seminar/Group Discussion(GD)

In Seminar or Group Discussion (GD) students have to select general topics related to current affairs or related to subject & make a PPT of the selected topic and present in front of the faculties & students. By doing this, in front of faculties & students, students communication skill will get developed & will be a good presenter. By doing this repetitively students subject knowledge will also get enhanced. Rather than making PPT, student can also discuss in the group of students in front of faculties that is called as Group Discussion.

